A LOSO and XAI Approach for Reliable Decision-Making under Spatial Uncertainty in Pasture Biomass Estimation

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Abstract

One of the most critical challenges in applying Deep Learning (DL) in agriculture is the lack of domain generalization for plant biomass estimation tasks. This study evaluates the robustness of a ResNet50 model using the CSIRO Image2Biomass dataset and a rigorous Leave-One-State-Out (LOSO) cross-validation protocol, designed to simulate deployment scenarios in unseen geographical domains. Although internal validation metrics indicate promising performance, the model fails catastrophically when predicting in regions such as New South Wales (a 106% error increase) and Western Australia (a 130% error increase), in contrast to its stability in Tasmania and Victoria. To diagnose the causes of these failures, Explainable Artificial Intelligence (XAI) techniques based on Grad-CAM were incorporated. The visual analysis revealed two distinct error patterns: a semantic failure, where the model focuses on vegetation but fails to quantify it, and a context failure, where the model relies on spurious soil correlations instead of biological features. These results demonstrate that standard validation metrics are insufficient to evaluate DL models in complex agricultural contexts. The combination of LOSO validation and XAI analysis offers a robust methodological framework to detect generalization failures, strengthening transparency and reproducibility in AI applications for sustainable agriculture. This study addresses the uncertainty inherent in agricultural AI models by diagnosing spatial generalization risks, providing a framework for intelligent decision-making regarding model reliability in unseen environments.

1 Introduction

In the context of precision agriculture, the integration of Deep Learning (DL) models with satellite and drone imagery has shown great potential for estimating agroecological variables such as plant biomass [1]. Various works report that DL models can surpass traditional methods by capturing complex relationships in remote data, improving, for example, agricultural yield prediction from satellite image time series and climate data[2].

Similarly, in biomass estimation, neural network-based approaches have achieved high levels of precision. A recent systematic review has highlighted that machine learning, including neural networks, is a key technique for optimizing accuracy in UAV-based pasture monitoring [3]. These deep models consistently surpass traditional algorithms; for instance,[4]—a study also using high-resolution UAV images, found their neural network model clearly outperformed Random Forest (RF) and Support Vector Machines (SVM). This finding is echoed by other comparisons, such as [5], who reported a median error (RMSE) approximately 50% lower for their neural network compared to RF/SVM models in pasture estimation. Consequently, high-precision metrics are consistently reported, with studies using advanced DL and sensor fusion achieving determination coefficients (R^2) near or exceeding 0.90 [6] or very low prediction errors [7], demonstrating the utility of these tools for large-scale agricultural management.

However, most of these models are often trained and evaluated under relatively homogeneous or controlled conditions, which does not adequately reflect real-world visual and ecological heterogeneity.

In practice, substantial variations exist between regions regarding soil type, vegetation species, foliage coloration, or cultivation patterns, which drastically alter the appearance of the input images. When a model trained in a limited context is operated in a geographical region unseen during training, such differences can cause a marked domain shift that severely hinders generalization. For example, previous research notes that discrepancies in illumination, plant phenological state, and soil type between different datasets produce a "domain shift" so pronounced that it is extremely challenging to extrapolate a model from one environment to another [8]. Furthermore, pasture systems present a diversity of species, structural complexity, and environmental variability much greater than monospecific crops, posing additional challenges for agricultural computer vision [1].

Consequently, a model may end up learning contextual cues specific to its training zone (e.g., soil tone or certain vegetation patterns) instead of visual features intrinsically linked to biomass, thus limiting its ability to generalize. In the absence of domain-oriented validation strategies, such as Leave-One-State-Out (LOSO), conventional metrics can offer a misleading view of the model's true extrapolation capability. Standard cross-validations (which mix data from all regions) tend to overestimate performance, as the model is evaluated on distributions similar to those of the training set. Conversely, by requiring the model to predict in a territory completely excluded from training (e.g., leaving a state or region out), sharp drops in performance frequently emerge. A recent study on agricultural yield prediction showed that, under a leave-one-region-out scheme, the model suffered drastic declines in precision, even obtaining negative R^2 values in the omitted region. Similarly, in cover crop biomass estimation, models calibrated with multi-region data barely reached R^2 values of 0.06–0.46 when tested on a completely new region, evidencing very limited spatial generalization despite their good results in conventional internal validations [9]. In summary, without implementing domain-based validation (such as the LOSO approach applied here), the robustness of DL models in this field can be easily overestimated.

Moreover, interpretability tools are rarely applied to understand why the model succeeds or fails under certain conditions. In the context of agricultural remote sensing, this opacity is aggravated by data complexity (multiple spectral bands, temporal variations, different resolutions), as well as factors like noise, shadows, or atmospheric conditions, which can obscure the model's internal decision-making process [10]. Consequently, explainability in DL models applied to crop imagery is crucial to ensure that humans, farmers, agronomists, or decision-makers, understand and trust the predictions before using them in the field [10]. Post-hoc techniques like Grad-CAM (Gradient-Weighted Class Activation Mapping) exist, which allow visualization of which image regions most influence the network's decision, thereby aligning the model's visual cues with expert knowledge of the phenomenon [11]. However, such explainability approaches have been uncommon in this domain, leaving a gap in understanding the reasons behind the model's predictions in different contexts.

This combination of insufficient validation and "black box" opacity introduce a critical barrier to adoption. In fields where management decisions, such as stocking rates or fertilization, carry economic and environmental stakes, farmers and agronomists require transparency to trust algorithmic outputs [12],

[13]. Explainability is, therefore, not an optional technical feature but a fundamental requirement for the auditing, accountability, and ethical deployment of precision agriculture systems [10], [14], [15].

In response to these identified gaps, this study evaluates the robustness and explainability of a Deep Learning model for biomass estimation against unseen geographical domains. To this end, a rigorous Leave-One-State-Out (LOSO) cross-validation protocol is employed to simulate spatial extrapolation scenarios, which is essential for data with spatial non-independence [16]. Complementarily, Grad-CAM-based interpretation techniques [17] are applied to visually identify the features the model uses, providing information on why specific decisions are made. It is hypothesized that the model, despite satisfactory internal metrics, will fail significantly when exposed to a new domain because of a tendency to learn spurious contextual patterns, such as soil texture or background noise, rather than biologically meaningful visual concepts [18]. The results obtained in this study confirm this hypothesis, revealing error increases of over 100% in unseen regions and demonstrating via XAI that the model engaged in "shortcut learning" by focusing on soil background rather than vegetation. While deep learning has shown promise in biomass estimation, the uncertainty of its performance across different geographical regions remains a major barrier to its use in intelligent decision-making systems. This work uses XAI and LOSO to quantify this uncertainty and ensures more robust application of computer vision models in agricultural infrastructure (SDG 9).

2 Materials and Methods

2.1 Dataset and Preparation

The CSIRO – Image2Biomass Prediction dataset, hosted on the Kaggle platform [1], was selected for this analysis. The original metadata file (train.csv) contained 1,785 records presented in a "long format," wherein each unique image was associated with five distinct rows corresponding to different biomass target variables. To align the data structure with the model's input requirements, a transformation to a "wide for-mat" was performed. This process consolidated the records into 357 unique samples, where each entry represents a single image linked to its metadata (e.g., State, Species, Pre_GSHH_NDVI) and the five regression targets: Dry_Clover_g, Dry_Dead_g, Dry_Green_g, Dry_Total_g, and GDM_g. An example of the dataset structure is presented in Fig. 1.

2.2 Exploratory Data Analysis (EDA) and Validation Strategy Dataset and Preparation

An Exploratory Data Analysis (EDA) was conducted to assess the dataset's structural characteristics. The 357 samples originate from four distinct Australian states (geographical domains): New South Wales (NSW), Tasmania (Tas), Victoria (Vic), and Western Australia (WA). However, the geographic distribution is highly imbalanced, as detailed in Table 1. Notably, the WA subset constitutes only 9% of the data (32 samples), presenting a challenge for model training.

Table 1
Distribution of samples by geographical domain (State).

State	Sample Size (n)
New South Wales (NSW)	75
Tasmania (Tas)	138
Victoria (Vic)	112
Western Australia (WA)	32
Total	**357**

Furthermore, biomass distributions vary significantly across these domains, as illustrated by the Dry_Total_g boxplot in Fig. 2. This variability indicates high data heterogeneity; visual features such as soil type, vegetation greenness, and pasture density are not consistent across state lines. This combination of domain imbalance and intrinsic heterogeneity limits the applicability of standard random k-fold cross-validation, which would likely lead to overly optimistic performance estimates. Consequently, a rigorous Leave-One-State-Out (LOSO) cross-validation protocol was adopted as the primary methodology. This approach is designed to simulate a realistic deployment scenario where the model must predict biomass in a geographical domain entirely unseen during the training phase.

2.3 Data Preprocessing and Augmentation

2.3.1 Data Preprocessing and Augmentation

To prepare images for evaluation (i.e., on the validation and test sets), a standard preprocessing pipeline was applied. This ensures that predictions are deterministic and reflect the model's true performance on unseen data. The steps are as follows:

1. *Resizing*: The images were resized to a uniform input resolution of 384 × 384 pixels. This resolution, higher than the 224 × 224 standard, was chosen to preserve fine-grained details of the pasture texture.
2. *Tensor Conversion*: The images were converted from PIL objects to PyTorch tensors, effectively scaling pixel values to the range [0, 1].
3. *ImageNet Normalization*: The tensor values (denoted as x) were normalized using the mean (μ) and standard deviation (σ) of the ImageNet dataset [19] to align the input distribution with the pre-trained weights of the ResNet50 network [20]. The normalization for each channel c (where $c \in \{R, G, B\}$) is applied according to Eq. 1:

$$\mathrm{x}'_{\mathrm{c}} = \frac{(\mathrm{x}_{\mathrm{c}} - \mu_{\mathrm{c}})}{\sigma_{\mathrm{c}}}$$

1

Where μ = [0.485, 0.456, 0.406] and σ = [0.229, 0.224, 0.225].

2.3.2 Training Set Data Augmentation

To combat overfitting and improve model robustness on the small training dataset (N = 357), a pipeline of online data augmentations was applied exclusively to the train-ing images. These transformations were applied randomly during training to create visual diversity:

- Geometric Augmentation: Random horizontal flips (p = 0.5) and random vertical flips (p = 0.5) were applied.
- Affine Transformation: Random affine transformations were used, including rotations (± 10 degrees), translation (± 10% in x/y), and scaling (± 10% zoom).
- Color Augmentation: ColorJitter was applied to randomly modify the image brightness (± 10%), contrast (± 10%), and saturation (± 10%).

This strategy forces the model to learn invariant features related to biomass, rather than memorizing superficial color tones or specific orientations from the training domains [21].

2.3.3 Training Set Data Augmentation

A critical aspect of the methodology was the handling of the output variables. Unlike the image inputs, the target variables were not preprocessed. The model was trained to predict the five biomass measurements (components and totals) directly in their raw values (e.g., grams), with a Softplus activation ensuring non-negative outputs. The logarithmic transformation was applied dynamically inside the custom loss function, WRMSLELoss. This metric was chosen as it is robust to outliers and optimizes for relative error.

The loss function was implemented as the sum of the individual Root Mean Squared Logarithmic Errors (RMSLE) for each of the K = 5 target components, weighted equally (w_i = 0.2). First, the component-wise RMSLE (RMSLEi) for each target i over a batch of N samples (where j is a sample, p is the prediction, and a is the actual value) is calculated according to Eq. 2:

$$RMSLE_i = \sqrt{\frac{1}{N}\ \sum\ _{j=1}^{N}\left(\log\left(p_{ij}+1\right)-\log\left(a_{ij}+1\right)\right)^2}$$

2

Finally, the total Weighted Loss (*Ltotal*) is computed as the weighted sum of these components, as defined in Eq. 3:

$$Loss_{WRMSLE} = \sum\ _{i=1}^{k=5} w_i \cdot RMSLE_i$$

3

This deferred processing strategy ensures that the model is directly optimized for the study's custom evaluation metric, minimizing the relative error rather than the absolute magnitude.

2.3.4 Model Architecture and Training Configuration

A ResNet50 architecture [20], pre-trained on the ImageNet dataset [19], was selected for this study. The choice of ResNet50, a model with high complexity (25.6M parameters), renders it particularly susceptible to overfitting on a small dataset (N = 357). This susceptibility underscores the critical need for a rigorous validation protocol like LOSO to diagnose true generalization failure, which standard validation metrics would otherwise conceal.

To adapt the model for this multi-output regression task, the final classification layer (2048 input features) was replaced with a custom regression head. This head consists of a linear layer reducing the features to 512, followed by a ReLU activation and a Dropout layer (p = 0.3) for regularization. A final linear layer maps these fea-tures to the five target outputs: Dry_Clover_g, Dry_Dead_g, Dry_Green_g, Dry_Total_g, and GDM_g. A Softplus activation function was applied to the final output layer to ensure all biomass predictions were non-negative.

For exclusively evaluating the learning capability from visual features, numerical metadata (e.g., NDVI or average height) were intentionally excluded from the training process. This methodological decision isolates the model's performance to be based purely on the spectral and structural information extracted from the RGB images.

The model was trained using the PyTorch framework [22] on a computational environment accelerated by two NVIDIA Tesla T4 GPUs. To enable parallel training, the network was wrapped in nn.DataParallel. The training hyperparameters were defined as follows:

- Optimizer: The AdamW optimizer [23] was used with learning rate (lr) of 1×10^{-4}.
- Loss Function: Instead of a standard Mean Squared Error (MSE), the custom WRMSLELoss (Weighted Root Mean Squared Logarithmic Error) function was implemented, as detailed in Eq. (3). This directly aligns the model's optimization with the study's evaluation metric.
- Batch Size: A batch size of 32 was utilized.
- Epochs and Model Selection: The model was trained for a maximum of 50 epochs. To prevent overfitting and select the optimal model, an Early Stopping mechanism was implemented. This mechanism monitored the loss on the internal validation set (WRMSLE_Val) after each epoch. If the validation loss did not improve for 5 consecutive epochs (i.e., patience = 5), the training was halted. The model weights from the epoch with the lowest validation loss were saved as the final model for evaluation on the test set.
- Reproducibility: A single global random seed of 42 was set for all libraries (NumPy, PyTorch, and Python's random) to ensure the full reproducibility of the data splitting, model initialization, and training processes.

2.3.5 Experimental Design

To quantify the model's robustness against unseen geographical domains, a "Leave-One-State-Out" (LOSO) cross-validation protocol was designed. This type of spatial block cross-validation is essential for data with spatial non-independence, as standard k-fold validation can produce overly optimistic performance estimates [16].

In this LOSO procedure, each of the four Australian states (NSW, Tas, Vic, and WA) was sequentially excluded from the training data and held out as an independent test set (unseen domain). The three remaining states constituted the training and validation set (trainval_df), from which an internal 20% validation split was used for model selection.

This approach simulates the model's deployment in entirely new geographical regions, permitting a realistic estimation of its inter-domain generalization. The entire LOSO procedure (4 folds) was executed using the single global random seed (42) spec-ified in Section 2.3.4 to ensure full reproducibility of the data splits and model initial-ization. This process is conceptually illustrated in Fig. 3.

2.3.6 Validation Process and Evaluation Metric

Within each of the 4 LOSO folds, a robust Best Model Saving strategy coupled with Early Stopping was applied. To perform internal model selection, the dataset from the seen domains (trainval_df) was randomly split into 80% for training and 20% for internal validation.

The model was trained for a maximum of 50 epochs. To prevent overfitting, the training process monitored the error on the internal validation set (WRMSLE_Val) at the end of each epoch. If the validation error did not improve for 5 consecutive epochs, training was automatically halted. Crucially, the model weights (best_model.pt) corresponding to the epoch with the lowest WRMSLE_Val were saved for the final evaluation.

This saved "best model" was then loaded for the final, single evaluation on the test set (the unseen state). The sole metric employed for model selection, optimization (via the WRMSLELoss function, Eq. (3)), and results reporting was the Weighted Root Mean Squared Logarithmic Error (WRMSLE). Using the same metric for both optimization and evaluation ensures a direct alignment between the training process and the study's core objective.

2.3.7 Interpretability Methodology (XAI)

To qualitatively understand the model's internal behavior and diagnose the causes of its generalization failures in the Western Australia (WA) and New South Wales (NSW) domains, an Explainable Artificial Intelligence (XAI) methodology based on Gradient-weighted Class Activation Mapping (Grad-CAM) [17] was implemented. This post-hoc technique allows for the visualization of the image regions that contribute most heavily to the model's final prediction.

In this study, Grad-CAM was applied to the last convolutional layer (layer4) of the ResNet50 architecture, utilizing the fold-specific optimal models (e.g., mod-el_fold_NSW.pt) saved during the LOSO protocol. The procedure consisted of calcu-lating the gradients of the regression output (for the specific target variable with the highest error) with respect to the activations of said layer, generating heatmaps that were overlaid onto the original images for analysis.

Images belonging to the WA and NSW domains that exhibited the highest pre-diction errors were specifically selected to explore the model's attention mechanisms in cases where its performance was deficient. The objective of the analysis was to determine whether the model's attention was focused on biologically relevant features (the vegetation) or if it relied on spurious correlations (such as background soil patterns). This post-hoc methodology complements the quantitative evaluation (Section 3.1), offering an interpretive tool to reinforce the transparency and reliability of the proposed diagnosis.

3 Results

3.1 Data Preprocessing and Augmentation

The results of the Leave-One-State-Out (LOSO) cross-validation are presented in Table 2. The ResNet50 model exhibited a significant domain generalization failure across all held-out states.

While the performance drop-off was notable in Tasmania (Tas) and Victoria (Vic)—with error increases of 1.50x and 1.33x respectively—the failure was cata-strophic when the model was evaluated on New South Wales (NSW) and Western Australia (WA). In these domains, the test error (WRMSLE) escalated dramatically, increasing by 2.06x (a 106% increase) and 2.30x (a 130% increase), respectively, compared to their internal validation scores.

On average, the LOSO protocol revealed a performance degradation of 80% (a 1.80x increase in error), with an average generalization error of 2.22, compared to an average internal validation error of 1.22Images belonging to the WA and NSW domains that exhibited the highest pre-diction errors were specifically selected to explore the model's attention mechanisms in cases where its performance was deficient. The objective of the analysis was to determine whether the model's attention was focused on biologically relevant features (the vegetation) or if it relied on spurious correlations (such as background soil patterns). This post-hoc methodology complements the quantitative evaluation (Section 3.1), offering an interpretive tool to reinforce the transparency and reliability of the proposed diagnosis.

Table 2
Average results of the LOSO protocol.

State (Test)	WRMSLE Val (Seen)	WRMSLE Test (Unseen)	Ratio (Test/Val)
NSW	1.24	2.56	2.06×
Tas	1.12	1.69	1.50×
Vic	1.20	1.59	1.33×
WA	1.33	3.05	2.30×
Average	1.22	2.22	1.80×

These results confirm the model's inability to generalize specific visual patterns learned from the training domains failing to maintain precision in visually distinct environments. Figure 4 summarizes this disparity, highlighting the performance gap between seen and unseen domains.

3.2 Component-wise Error Analysis

The analysis of individual biomass components (Fig. 5) reveals the specific nature of these failures. In the NSW domain, the largest contribution to the error came from the Clover_E and Green_E fractions. This suggests that the model struggled to recognize the visual patterns associated with both the specific legume species and the active green vegetation in that region.

In contrast, in WA, the error profile was distinct. The highest deviation was ob-served in the Dead_E fraction, although Clover_E and Green_E also exhibited substantial errors. These findings reflect that the failure is not uniform but is instead dependent on the dominant visual characteristics of each region, such as the vegetation mix in NSW or the complex background interactions (dead biomass and soil) in WA.

3.3 Visual Diagnosis with Grad-CAM

To diagnose the causes of the failures identified in Section 3.1, Grad-CAM was applied to the images corresponding to the highest component error from the NSW and WA domains.

The analysis of the NSW fold (Fig. 6a) targets the image ID384648061.jpg, which presented the largest Green_E error. For this sample, the model predicted only 26.92 g of green biomass, whereas the ground-truth value was 157.98 g (an absolute error of 131.05 g). The resulting activation map reveals a profound semantic failure. Instead of focusing on the abundant green vegetation, the model’s attention is diffuse and spread uniformly across the entire image. This "attentional wash" indicates that the model was completely confused and unable to identify the relevant visual features for "green" biomass in this new domain.

Conversely, the analysis of the WA fold (Fig. 6b) targets image ID1139918758.jpg. While Dead_E was the dominant error source in this state (as noted in Section 3.2), this specific sample was selected to illustrate a critical Clover_E failure (predicting 6.00 g vs. a real value of 58.34 g). The Grad-CAM reveals a clear context failure. The model's attention is not dispersed; rather, it is surgically focused on the textural patterns of the dry, cracked soil. This demonstrates that the model incorrectly learned a spurious correlation between this specific soil texture and the biomass, causing it to completely ignore the actual clover in the image.

4 Results

The quantitative (Table 2, Figs. 4 and 5) and qualitative (Fig. 6) results empirically validate the scientific hypothesis: Deep Learning models based solely on visual features tend to depend on local contextual correlations, limiting their spatial generalization. This finding aligns with previous research highlighting the vulnerability of agricultural models to domain shifts caused by environmental variability [8].

The use of XAI (Grad-CAM) allows moving beyond numerical observation to causal interpretation, revealing the model's internal attention mechanisms and diagnosing the different forms of "model blindness" observed in the failed domains. This qualitative analysis provides visual evidence confirming that the model does not always learn robust biological concepts, but rather textural and contextual patterns associated with the training environments, reinforcing the argument for explainability as a mandatory auditing tool [10], [15].

4.1 NSW Case: Interpretation of Semantic Failure

In the New South Wales (NSW) domain, the Clover_E and Green_E biomass components exhibited the highest errors (Fig. 5). The Grad-CAM analysis (Fig. 6a) reveals a profound semantic failure. Unlike a robust model that would isolate the plant matter, the attention in NSW is characterized by a diffuse "attentional wash," spread uniformly across the entire image.

This behavior indicates that the model fails to conceptually recognize the "green vegetation" in this new domain. In cognitive terms, the network lacks a stable semantic representation of the biological object. Instead of identifying the plants, it treats the entire scene as a confusing input, likely because the visual features of NSW (e.g., spe-cific vegetation tones or planting arrangements) deviate significantly from the features learned in Tasmania and Victoria. Consequently, the model cannot "anchor" its pre-diction on the relevant biological pixels, leading to large estimation errors. This high-lights the limitations of models trained on visually homogeneous datasets, as they fail to capture the invariant features necessary for broad generalization [1].

4.2 Case WA: Interpretation of Context Failure (Shortcut Learning)

The Western Australia (WA) domain presents a distinct and more alarming error pattern. While Dead_E was the dominant error quantitatively, the qualitative analysis of the Clover_E failure (Fig. 6b) exposes the root cause: shortcut learning.

In contrast to the diffuse attention in NSW, the Grad-CAM maps in WA show attention that is incorrectly but strongly focused on the background, specifically, the texture of the dry, cracked soil. This suggests that during training, the model learned a spurious correlation (e.g., "biomass relates to specific background contrasts"). When applied to WA, where the soil context is visually distinct (lighter and sandier), this spurious rule misleads the model. It focuses surgically on the abiotic soil features rather than the sparse vegetation. This confirms that the model is not measuring biomass directly but is instead relying on superficial contextual cues that do not generalize, a phenomenon famously known in machine learning as "Clever Hans" behavior [18].

5 Conclusions

The quantitative (LOSO) and qualitative (XAI) findings of this study converge on a central conclusion: traditional internal validation metrics are insufficient to assess the reliability of Deep Learning models in real-world agricultural contexts. It has been empirically demonstrated that a vision model can exhibit solid internal validation metrics (as observed in the Tas and Vic domains) but fail drastically when deployed in a domain with different visual conditions. This was evidenced by the catastrophic performance degradation, with error increases of 106% (2.06x) in NSW and 130% (2.30x) in WA compared to validation scores.

The use of Explainable Artificial Intelligence (XAI) was crucial, as it allowed a shift from numerical observation to causal interpretation. The Grad-CAM analysis provided visual evidence that the model did not learn robust biological concepts but rather de-pended on spurious correlations that collapsed outside the training distribution.

In synthesis, the combination of a robust evaluation (LOSO protocol) and an interpretability analysis (Grad-CAM) constitutes a comprehensive methodological framework. This approach makes it possible to move beyond merely reporting an error and instead diagnose the specific mechanisms of failure, thereby improving the model's transparency and reliability before its deployment in sustainable agriculture applications.

By identifying shortcut learning through Grad-CAM, we reduce the uncertainty of model predictions, allowing farmers and stakeholders to make informed decisions based on reliable vegetation features rather than spurious background correlations.

5.1 Future Work

This approach not only identifies vulnerable domains but also guides future improvement strategies. Based on the XAI diagnostics, future work should focus on:

1. XAI-Informed Data Augmentation: Using the Grad-CAM findings (e.g., the context failure on light-colored soils in WA or the distinct vegetation patterns in NSW) to design targeted augmentation strategies that explicitly simulate the failure contexts during training.

2. Domain Generalization Techniques: Implementing advanced architectures, such as Domain-Adversarial Neural Networks (DANN) [24], that explicitly penalize the model for learning domain-specific features. This aligns with broader strategies in Domain Generalization [25] designed to force models to learn more invariant biological representations.
3. Multimodal Fusion: Investigating whether the inclusion of numerical metadata (e.g., NDVI, crop height), which were intentionally omitted in this visual spectrum study, can help the model resolve the visual ambiguity that caused the observed semantic and context failures.

Declarations

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Ethics approval

Ethics declaration: not applicable. (This study does not involve human participants or animals).

Consent to participate

Consent to Participate declaration: not applicable.

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Author Contribution

J.R.H.H. conceived the study, developed the LOSO and XAI pipeline, and wrote the main manuscript. R.D.R.C. and R.O.B.J. contributed to data analysis and methodology validation. All authors reviewed and approved the final manuscript.

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Data Availability

Publicly available datasets were analyzed in this study. This data (CSIRO - Image2Biomass Prediction) can be found here: https://www.kaggle.com/competitions/csiro-biomass. Reference number [1]. The source code used to conduct the experiments and generate the results for this study is openly available in Zenodo at: https://zenodo.org/records/17634972. Reference number [26].

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Figures

Figure 1

Example training samples from the CSIRO dataset [17] by domain. The images are rotated (90°) for visualization and paired with their corresponding metadata. The data highlights the high visual and quantitative variability between domains, including differences in species, NDVI, average height, and ground-truth biomass composition (g).

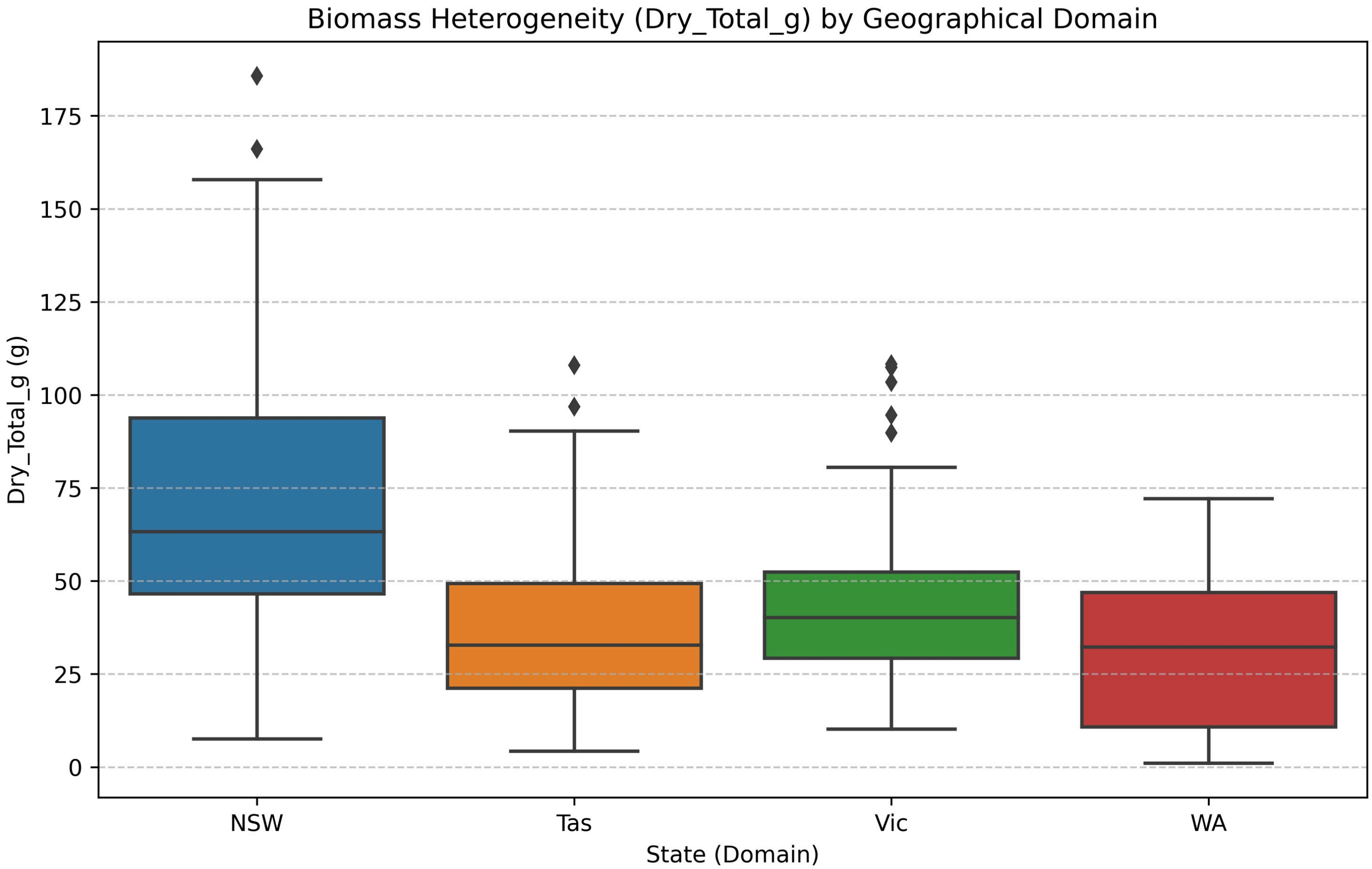

Figure 2

Exploratory analysis of biomass heterogeneity (Dry_Total_g) across the four geographical do-mains. The difference in medians and distributions justifies the LOSO methodology.

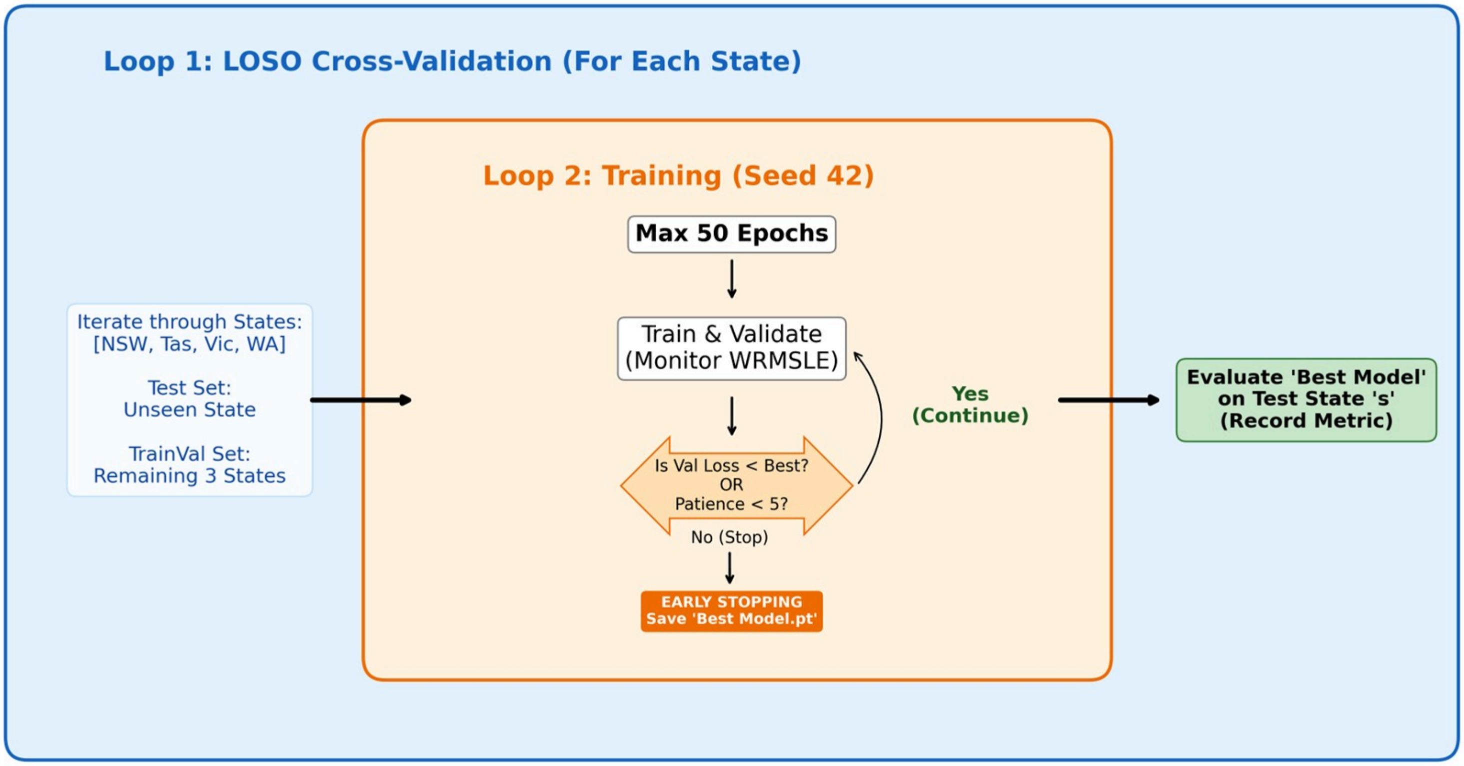

Figure 3

Conceptual diagram of the LOSO experimental design. (Loop 1) The outer loop iterates through each state (LOSO). (Loop 2) The inner loop trains the model (max 50 epochs), saving the best model based on internal validation (EarlyStopping).

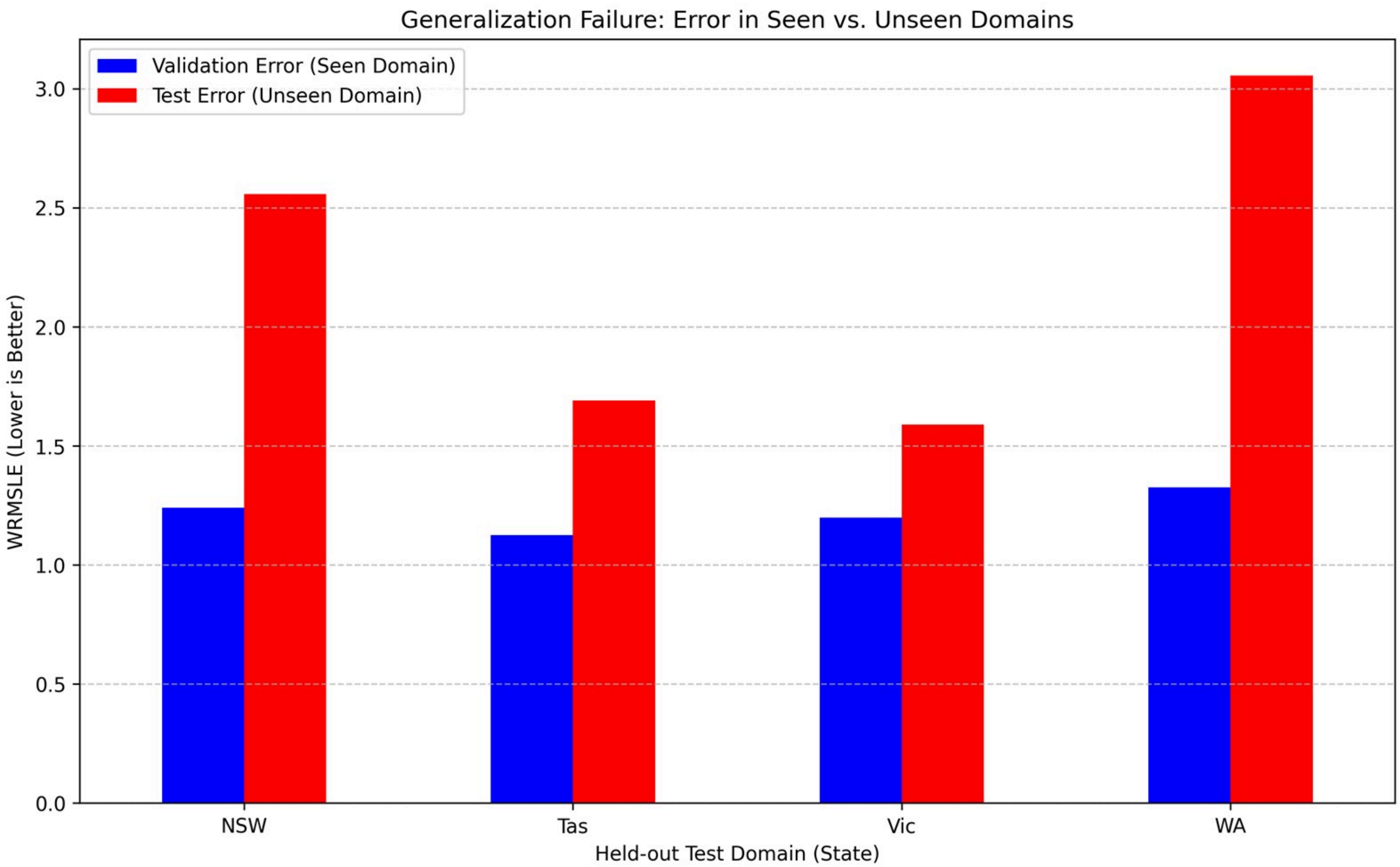

Figure 4

Performance gap analysis. The chart illustrates the sharp increase in error (WRMSLE) when the model is applied to unseen domains, particularly in NSW and WA.

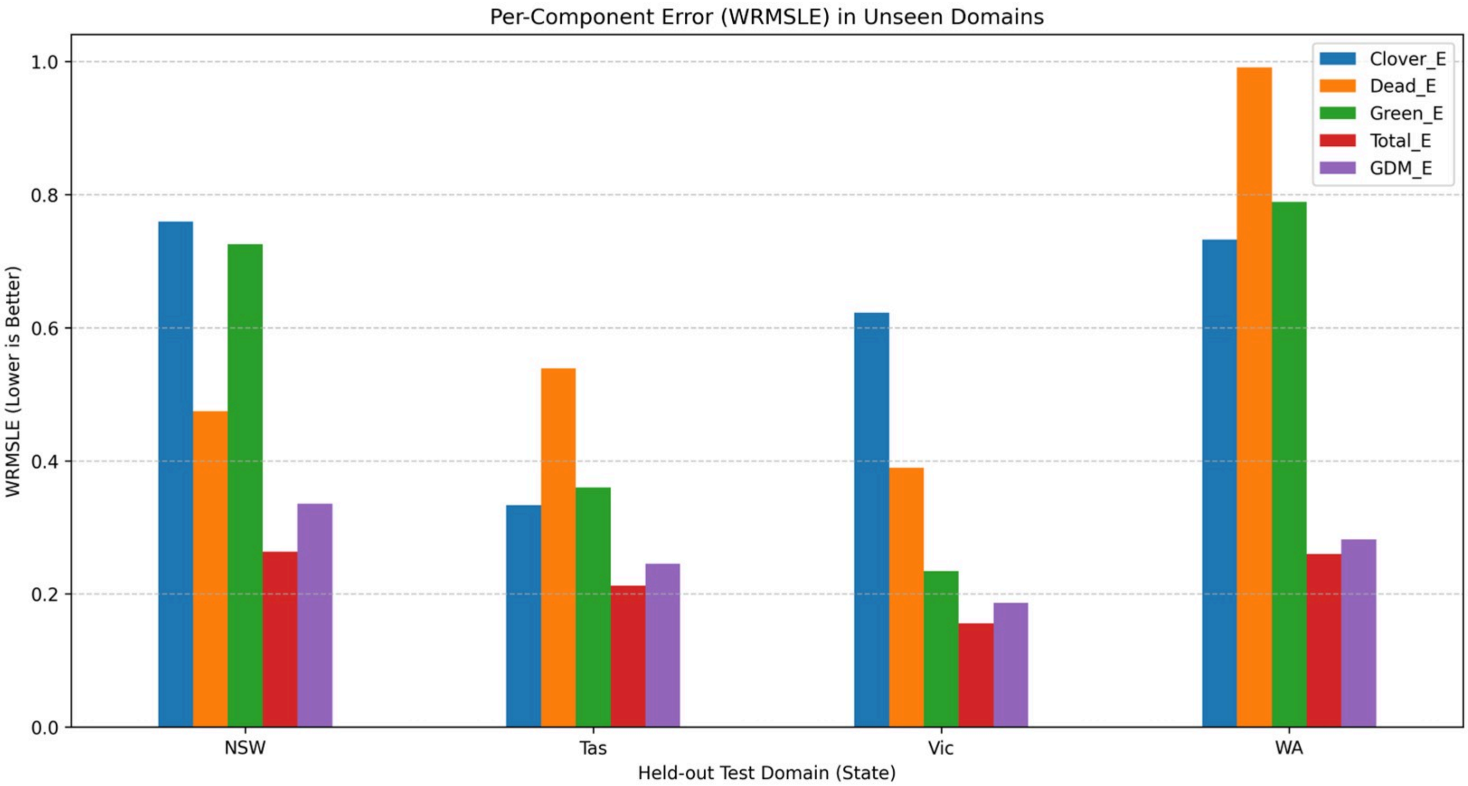

Figure 5

Breakdown of the test error (WRMSLE) by biomass component in each unseen domain. The _E suffix denotes the calculated error for each respective component.

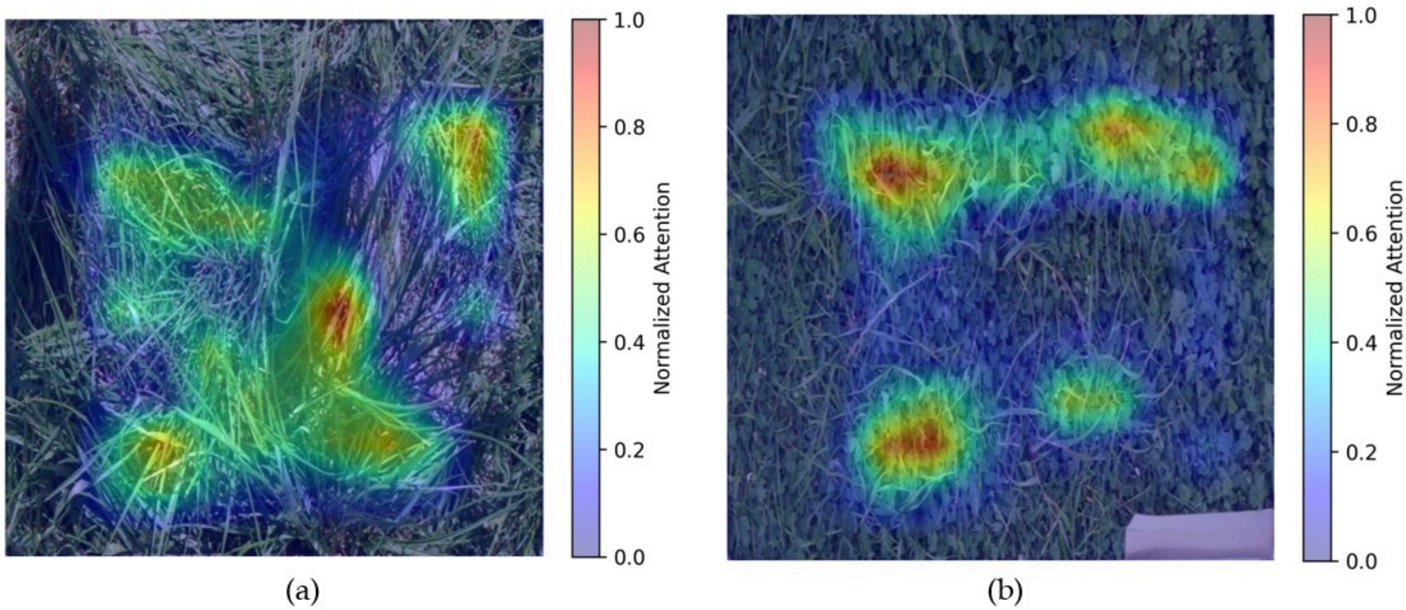

Figure 6

Visual diagnosis of generalization failures using Grad-CAM. (a) Semantic Failure in NSW: The activation map shows diffuse attention ("attentional wash") across the scene, indicating the model fails to conceptually isolate the green biomass; (b) Context Failure in WA: The model exhibits shortcut learning, focusing surgically on the background soil texture rather than the target clover vegetation.